25

$$\int_{-\infty}^{\infty} f(x)\,\delta'(x-a)\,dx = f(x)\,\delta(x-a)\Big|_{-\infty}^{\infty} - \int_{-\infty}^{\infty} f'(x)\,\delta(x-a)\,dx = -f'(a)$$

a) $$\tilde{g}_{cont} = \begin{cases} g_1(x) & \text{for } x<a \\ g_2(x)+G_0 & \text{for } x>a \end{cases} \qquad G_0 = \lim_{x\to -a} g_1(x) - \lim_{x\to +a} g_2(x) = g_1(a) - g_2(a)$$

$$D_x\, g(x) = D_x\, \tilde{g}_{cont} - G_0\,\delta(x-a) = \left\{ \begin{matrix} g_1'(x) & (x<a) \\ g_2'(x) & (x>a) \end{matrix} \right\} - G_0\,\delta(x-a)$$

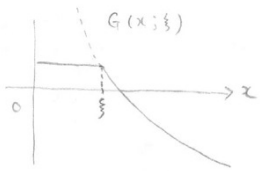
$$D_x^2\, g(x) = D_x\, g'(x) - G_0\, D_x\,\delta(x-a) = \left\{ \begin{matrix} g_1''(x) & (x<a) \\ g_2''(x) & (x>a) \end{matrix} \right\} - G_1\,\delta(x-a) - G_0\, D_x\,\delta(x-a)$$

$$D_x\,(D_x^n\, g(x)) = D_x\, g^{(n)}(x) - \sum_{j=0}^{n-1} G_j\, D_x^{n-j}\,\delta(x-a)$$

$$= g^{(n+1)}(x) - G_n\,\delta(x-a) - \sum_{j=0}^{n-1} G_j\, D_x^{n-j}\,\delta(x-a)$$

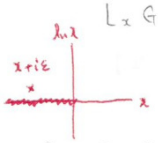
b) $$D_x\, G(x;\xi) = \frac{\partial}{\partial x} G(x;\xi)$$

$$D_x^2\, G(x;\xi) = \frac{\partial^2}{\partial x^2} G(x;\xi) - \left(0 - \left(-\frac{1}{\xi}\right)\right)\delta(x-\xi)$$



$$G' = \begin{cases} 0 \\ \frac{1}{x} \end{cases} \qquad G'' = \begin{cases} 0 \\ -\frac{1}{x^2} \end{cases}$$

$$L_x\, G(x;\xi) = (x\,D_x^2 + D_x)\,G(x;\xi) = \underbrace{x\,G''(x;\xi) + G'(x;\xi)}_{0} - \frac{x}{\xi}\,\delta(x-\xi) = -\delta(x-\xi)$$



c) $$D_x(\ln x) = \frac{1}{x+i\varepsilon} = P\frac{1}{x} - i\pi\,\delta(x)$$: Dirac の公式

principal value

$$= \frac{x}{x^2+\varepsilon^2} - \frac{i\varepsilon}{x^2+\varepsilon^2}$$

$$\ln x = \begin{cases} \ln x & (x>0) \\ \ln|x| + i\pi & (x<0) \end{cases}$$

d) $$D_x|x| = \begin{cases} 1 & (x>0) \\ -1 & (x<0) \end{cases}, \qquad D_x^2|x| = -2\,\delta(x) \qquad |x| = x\,\theta(x) - x\,\theta(-x)$$

$$= \theta(x) - \theta(-x)$$

e) $$f(x) = \frac{1}{x\pm i\varepsilon} = P\frac{1}{x} \mp i\pi\,\delta(x) \qquad \left(\frac{\partial}{\partial x}\right)^n \frac{1}{x\pm i\varepsilon} = (-1)^n n!\,\frac{1}{(x\pm i\varepsilon)^{n+1}}$$

$$D_x f(x) = -P\frac{1}{x^2} \mp i\pi\,\delta'(x)$$

$$D_x^{(n)} f(x) = (-1)^n n!\,P\frac{1}{x^{n+1}} \mp i\pi\,\delta^{(n)}(x)$$

~~25~~

$$\frac{1}{x+i\varepsilon} = \mathcal{P}\frac{1}{x} - i\pi\delta(x)$$

$$\int_{-\infty}^{\infty} dx\, f^{(n)}(x)\frac{1}{x+i\varepsilon} = \mathcal{P}\int_{-\infty}^{\infty} dx\, f^{(n)}(x)\frac{1}{x} - i\pi\int_{-\infty}^{\infty} dx\, f^{(n)}(x)\delta(x)$$

26. $L_x = a(x)D_x^2 + b(x)D_x + c(x) = D_x(p(x)D_x) + q(x)$

$$\begin{cases} L_x y(x) = f(x) \\ y(0) = y(a) = 0 \end{cases} \Rightarrow y(x) = \int_0^a \underbrace{G(x;\xi)}_{\text{Green 関数：連続 関数}} f(\xi)\,d\xi$$

$$\begin{cases} L_x G(x;\xi) = \delta(x-\xi) \\ G(0;\xi) = G(a;\xi) = 0 \\ 0 < \xi < a \end{cases}$$

$$G(x;\xi)|_{x=\xi+0} = G(x;\xi)|_{x=\xi-0}$$

$$\int_{\xi-\varepsilon}^{\xi+\varepsilon} L_x G(x;\xi)\,dx = \int_{\xi-\varepsilon}^{\xi+\varepsilon}\delta(x-\xi)dx = 1$$

$$D_x G(x;\xi)|_{x=\xi+0} - D_x G(x;\xi)|_{x=\xi-0} = \frac{1}{p(x)}$$

$G(x;\xi) = G(\xi;x)$ 対称

a) $D_x^2 G(x;\xi) = \delta(x-\xi)$

$0 \le x < \xi \quad G(x;\xi) = \alpha x$

$\xi < x \le a \quad G(x;\xi) = \beta(x-a)$

連続性 $\alpha\xi = \beta(\xi-a)$

微分のとび $\beta - \alpha = 1$

$$\begin{cases} \alpha = \frac{\xi}{a} - 1 \\ \beta = \frac{\xi}{a} \end{cases} \rightarrow G(x;\xi) = \begin{cases} \frac{x}{a}(\xi-a) \\ \frac{\xi}{a}(x-a) \end{cases}$$

$$G(x;\xi) = \frac{x}{a}(\xi-a)\theta(\xi-x) + \frac{\xi}{a}(x-a)\theta(x-\xi) \Rightarrow G(x;\xi) = G(\xi;x)$$

odd ↓ even ↓

$x\delta(x) = 0$

$x\delta'(x) = -\delta(x)$

b) $L_x = D_x(xD_x) - \frac{1}{x} \quad u(x) = x^\lambda \rightarrow (\lambda^2-1)x^{-\lambda-1} = 0 \quad \lambda = \pm 1$

$$G(x;\xi) = \begin{cases} \alpha x \\ \beta\left(x - \frac{1}{x}\right) \end{cases}$$

連続性 $\alpha\xi = \beta\left(\xi - \frac{1}{\xi}\right)$

微分のとび $\beta\left(1 + \frac{1}{\xi^2}\right) - \alpha = \frac{1}{\xi}$

$$\begin{cases} \alpha = \frac{1}{2}\left(\xi - \frac{1}{\xi}\right) \\ \beta = \frac{1}{2}\xi \end{cases} \rightarrow G(x;\xi) = \begin{cases} \frac{x}{2}\left(\xi - \frac{1}{\xi}\right) \\ \frac{\xi}{2}\left(x - \frac{1}{x}\right) \end{cases}$$

$$G(x;\xi) = \frac{x}{2}\left(\xi - \frac{1}{\xi}\right)\theta(\xi-x) + \frac{\xi}{2}\left(x - \frac{1}{x}\right)\theta(x-\xi)$$

e) $L_x = D_x^2 - \lambda^2$

$$G(x;\xi) = \begin{cases} A\,\mathrm{sh}\,\lambda x \\ AK\,\mathrm{sh}\,\lambda(x-a) \end{cases}$$

$$\mathrm{sh}\,\lambda\xi = K\,\mathrm{sh}\,\lambda(\xi-a)$$

$$A\lambda\cos\lambda\xi - AK\lambda\cos\lambda(\xi-a) = 1$$

$$G(x;\xi) = \begin{cases} A\,\mathrm{sh}(\lambda x)\,\mathrm{sh}(\lambda(\xi-a)) & x < \xi \\ A\,\mathrm{sh}(\lambda\xi)\,\mathrm{sh}(\lambda(x-a)) & x > \xi \end{cases}$$

=

26.

$$\int_{\xi-0}^{\xi+0} D_x^2 G(x;\xi)\,dx = \int_{\xi-0}^{\xi+0} \delta(x-\xi)\,dx \quad (=1)$$

a) $D_x^2 G(x;\xi) = \delta(x-\xi) \rightarrow D_x G(x;\xi) = \theta(x-\xi) + C(\xi)$

$D_x G(\xi+0;\xi) - D_x G(\xi-0;\xi) = \theta(+0) - \theta(-0) = 1$

$L_x u(x) = 0 \rightarrow u(x) = Ax + B$

$L_x G(x;\xi) = G''(x;\xi) - G'_0 D_x \delta(x-\xi) - (-1)\,\delta(x-\xi)$

$\begin{cases} 0<x<\xi & G(x;\xi) = Ax + B \\ \xi<x<a & G(x;\xi) = A'x + B' \end{cases}$

$D_x G(x;\xi) = A + (A'-A)\theta(x-\xi)$ ($A'-A$: 1)

$L_x G(x;\xi) = (A'-A)\delta(x-\xi)$

$G(x;\xi) = \begin{cases} (\frac{\xi}{a}-1)x & (0\le x<\xi) \\ \frac{\xi}{a}x - \xi & (\xi<x\le a) \end{cases}$

b) $L_x u(x) = \left\{D_x(xD_x) - \frac{1}{x}\right\}u(x) = 0 \qquad u(x) = Ax + \frac{B}{x}$

$G(0;\xi) = G(1;\xi) = 0$ より $G(x;\xi) = \begin{cases} Ax & (0\le x<\xi) \\ A'(x-\frac{1}{x}) & (\xi<x\le 1) \end{cases} \rightarrow G(x;\xi) = \begin{cases} (\xi-\frac{1}{\xi})\frac{x}{2} & (0\le x<\xi) \\ \xi(x-\frac{1}{x})\frac{1}{2} & (\xi<x\le 1) \end{cases}$

c) $L_x G(x;\xi) = \delta(x-\xi) \rightarrow L_x \int G(x;\xi) f(\xi)\,d\xi = \int f(\xi)\delta(x-\xi)\,d\xi = f(x)$

$\therefore y(x) = \int G(x;\xi) f(\xi)\,d\xi + z(x)$

$L_x\left(y(x) - \int G f\,d\xi\right) = 0$

$L_x z(x) = 0$

$z(0) = z(1) = 0$

$= \frac{x-a}{a}\int_0^x \xi f(\xi)\,d\xi + \frac{x}{a}\int_x^a (\xi-a) f(\xi)\,d\xi \qquad \leftarrow y(0) = y(a) = 0$

d) $y(x) = \int G(x;\xi) f(\xi)\,d\xi = \int_0^x (\frac{\xi}{a}-1)x\, f(\xi)\,d\xi + \int_x^a (\frac{x}{a}-1)\xi f(\xi)\,d\xi$

$= \frac{x}{a}\int_0^a \xi f(\xi)\,d\xi - x\int_0^\xi f(\xi)\,d\xi - \underbrace{\int_\xi^a \xi f(\xi)\,d\xi}_{\text{"}0}$

27.

a) $G_\lambda = \sum_n c_n u_n(x)$

$(L_x - \lambda) G_\lambda = \sum_n (\lambda_n - \lambda) c_n u_n(x) = \delta(x-\xi)/\omega(x)$

両辺に $\bar{u}_m \omega(x)$ をかけて積分

$$\sum_n (\lambda_n - \lambda) c_n \underbrace{\int_0^a \bar{u}_m u_n \omega \, dx}_{\delta_{mn}} = \int_0^a \delta(x-\xi)\bar{u}_m(x)\,dx$$

δ_{mn} → $(\lambda_m - \lambda) c_m$; $\bar{u}_m(\xi)$

$$\rightarrow c_m = \frac{\bar{u}_m(\xi)}{\lambda_m - \lambda}$$

b) $\frac{1}{2\pi i}\int_C G_\lambda \, d\lambda = \sum_n \frac{1}{2\pi i}\int_C \frac{\bar{u}_n(\xi) u_n(x)}{\lambda_n - \lambda} d\lambda = -\sum_n \bar{u}_n(\xi) u_n(x) = -\sum_n \frac{\bar{u}_n(\xi)}{\lambda_n - \lambda}(\lambda_n - \lambda) u_n(x)$

$= -\sum_n \frac{\bar{u}_n(\xi)}{\lambda_n - \lambda}(L_x - \lambda) u_n(x) = -(L_x - \lambda)\underbrace{\sum_n \frac{\bar{u}_n(\xi) u_n(x)}{\lambda_n - \lambda}}_{G_\lambda} = -\delta(x-\xi)/\omega(x)$

c) $L_x u_n = D_x^2 u_n = \lambda_n u_n$

$u_n(0) = u_n(a) = 0 \rightarrow \lambda_n < 0 \qquad u_n = c_n \sin(\sqrt{\lambda_n}\, x)$

$\int_0^a \sin^2\left(n\pi \frac{x}{a}\right) dx = \frac{a}{2}$

$\sqrt{|\lambda_n|}\, a = n\pi$ (n: 整数) $\qquad \lambda_n = \left(\frac{n\pi}{a}\right)^2$

規格化して $c_n = \sqrt{\frac{2}{a}}$ $\qquad G_\lambda = \textcircled{?} \sum_{n=1}^{\infty} \frac{\frac{2}{a}\sin\left(\frac{n\pi}{a}x\right)\sin\left(\frac{n\pi}{a}\xi\right)}{-\left(\frac{n\pi}{a}\right)^2 - \lambda^2}$

28

a) $\int a(\vec{k})\, F(ik_i)\, e^{i\vec{k}(\vec{r}-\vec{r}')}\, d\vec{k} = \delta(\vec{r}-\vec{r}') = \frac{1}{(2\pi)^n}\int e^{i\vec{k}(\vec{r}-\vec{r}')}\, d\vec{k}$

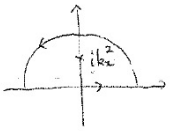
$F(D_{x_i})$ ↑ $k_1(x_1-x_1')+k_2(x_2-x_2')+\cdots$

$\therefore\ a(\vec{k}) = \dfrac{1}{(2\pi)^n F(ik_i)} \rightarrow G_s(\vec{r},\vec{r}') = \dfrac{1}{(2\pi)^n}\int \dfrac{e^{i\vec{k}(\vec{r}-\vec{r}')}}{F(ik_i)}\, d\vec{k}$

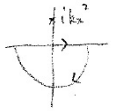
$\left(d\vec{k} = dk_1 \cdot dk_2 \cdots \cdot dk_n \right)$

b) $F = (ik_x)^2 - ik_y$

$G_s = \dfrac{1}{4\pi^2}\int \dfrac{e^{i\{k_x(x-x')+k_y(y-y')\}}}{-k_x^2 - ik_y}\, dk_x dk_y = \dfrac{i}{4\pi^2}\int dk_x\, e^{ik_x(x-x')} \underbrace{\int dk_y \dfrac{e^{ik_y(y-y')}}{k_y - ik_x^2}}_{I}$



$I = \begin{cases} 2\pi i\, e^{i\cdot ik_x^2(y-y')} = 2\pi i\, e^{-k_x^2(y-y')} \\ 0 \quad (y<y') \end{cases}$



$(y>y')\quad G_{Ts} = -\dfrac{1}{2\pi}\int dk_x\, e^{ik_x(x-x')-k_x^2(y-y')} = \dfrac{-1}{2\pi}\sqrt{\dfrac{\pi}{y-y'}}\, e^{-\frac{(x-x')^2}{4|y-y'|}}$

c) $L_{x,y} = D_x^2 - D_y^2 \qquad (y = k^2 t)$

$u(x,t) = \int G(x,y;x',y')\, f(x')\, dx'$

$= -\dfrac{1}{\sqrt{4\pi k^2 t}}\int f(x')\, e^{-\frac{(x-x')^2}{4k^2 t}}\, dx'$

$G_s(x,t,x',t') = -\theta(t-t')\dfrac{1}{\sqrt{4\pi k^2|t-t'|}}\, e^{-\frac{(x-x')^2}{4k^2|t-t'|}}$

$u(x,t) = \int_{-\infty}^{\infty} dx'dt'\, G_s(x,t,x',t')\, f(x')\, \delta(t')$

$= \theta(t)\dfrac{1}{2k\sqrt{t\pi}}\int_{-\infty}^{\infty} dx'\, e^{-\frac{(x-x')^2}{4k^2 t}}\, f(x')$

$\dfrac{\partial^2 u}{\partial x^2} - \dfrac{1}{k^2}\dfrac{\partial u}{\partial t} = 0 \qquad (t>0)$

初期条件
- $u = 0 \quad (t<0)$
- $u = f(x) \quad (t=0)$
- $u = ? \quad (t>0)$

拡張

$\dfrac{\partial^2 u}{\partial x^2} - \dfrac{1}{k^2}\dfrac{\partial u}{\partial t} = \delta(t)\, g(x)$

$t = -\varepsilon \sim \varepsilon$ で積分する → $g(x)$ が決まる

$\int_{-\varepsilon}^{\varepsilon} dt \dfrac{\partial^2 u}{\partial x^2} - \dfrac{1}{k^2}\int_{-\varepsilon}^{\varepsilon} dt \dfrac{\partial u}{\partial t} = \underbrace{\int_{-\varepsilon}^{\varepsilon} dt\, \delta(t)\, g(x)}_{1}$

$-\dfrac{1}{k^2}(u(x,\varepsilon) - u(x,-\varepsilon))$

↓ ↓

$-\dfrac{1}{k^2} f(x) \cdot g(x)$

$\dfrac{\partial^2 u}{\partial x^2} - \dfrac{1}{k^2}\dfrac{\partial u}{\partial t} = -\dfrac{1}{k^2} f(x)\, \delta(t)$

と Green 関数を使って解く

28

c) $t \to 0$ の ときの を調べる ために

$$\delta(x) = \lim_{\alpha\to\infty} \alpha f(\alpha x) \Big/ \int_{-\infty}^{\infty} f(x)dx$$

(Proof) $(\alpha > 0)$

$$\int_{-\infty}^{\infty} \alpha f(\alpha x) g(x) dx = \int_{-\infty}^{\infty} f(y)\, g\left(\frac{y}{\alpha}\right) dy \xrightarrow[\alpha\to\infty]{} g(0)\int_{-\infty}^{\infty} f(y)dy \qquad \text{c.f. } \int f(x)\delta(x-a)dx = f(a)$$

ここで $f(x) = e^{-x^2}$ を 利用すると $\int_{-\infty}^{\infty} f(x)dx = \sqrt{\pi}$ $\quad \lim_{\alpha\to\infty} \frac{1}{\sqrt{\pi}} \alpha e^{-\alpha^2 x^2} = \delta(x)$

$\alpha = \frac{1}{2k\sqrt{\pi t}} \xrightarrow{t\to 0} \infty$ より $\quad \lim_{t\to 0} \frac{1}{2k\sqrt{\pi t}} e^{-\frac{x^2}{4k^2 t}} = \delta(x)$

e) $G(r_x, r_y) = \frac{1}{(2\pi)^2}\int_{-\infty}^{\infty} dk_y \int_{-\infty}^{\infty} dk_x \frac{e^{i\vec{k}\cdot(\vec{r}-\vec{r}')}}{k_x^2 - k_y^2}$ $\quad r_x = x - x'$ $\quad \vec{r} = (x, y)$ $\quad r_y = y - y'$ $\quad \vec{r}' = (x', y')$

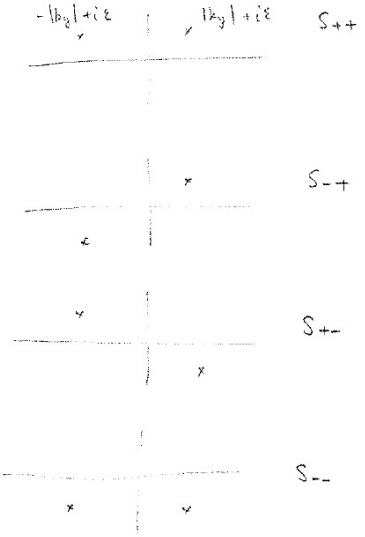
$$= \frac{1}{2\pi}\int_{-\infty}^{\infty} dk_y\, e^{ik_y r_y} \cdot \underbrace{\frac{1}{2\pi i}\int_{-\infty}^{\infty} dk_x \frac{e^{ik_x r_x}}{(k_x + |k_y|)(k_x - |k_y|)}}_{I}$$

I

$r_y > 0$ $\quad$ 上半円 $\quad \Rightarrow \theta(r_y)$

$r_y < 0$ $\quad$ 下半円 $\quad \Rightarrow \theta(-r_y)$

	$f(r_x)$	$\theta(-r_x)$
S_{++}	$\frac{e^{i\lvert k_y\rvert r_x}}{2\lvert k_y\rvert + i\varepsilon} + \frac{e^{i\lvert k_y\rvert r_x}}{-2\lvert k_y\rvert + i\varepsilon}$	0
S_{-+}	$\frac{e^{i\lvert k_y\rvert r_x}}{2\lvert k_y\rvert + i\varepsilon}$	$-\frac{e^{-i\lvert k_y\rvert r_x}}{2\lvert k_y\rvert - i\varepsilon}$
S_{+-}	$\frac{e^{i\lvert k_y\rvert r_x}}{-2\lvert k_y\rvert + i\varepsilon}$	$-\frac{e^{i\lvert k_y\rvert r_x}}{2\lvert k_y\rvert - i\varepsilon}$
S_{--}	0	$-\frac{e^{i\lvert k_y\rvert r_x}}{2\lvert k_y\rvert - i\varepsilon} - \frac{e^{-i\lvert k_y\rvert r_x}}{2\lvert k_y\rvert - i\varepsilon}$



S_{x+}

$$G = \frac{i}{2\pi}\int_{-\infty}^{\infty} dk_y\, e^{ik_y r_y}\left(\frac{e^{i|k_y|r_x}}{2|k_y|+i\varepsilon} + \frac{e^{-i|k_y|r_x}}{-2|k_y|+i\varepsilon}\right)\theta(r_x)$$

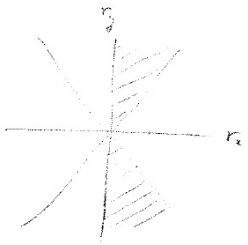
$$= \frac{i}{2\pi}\left\{\int_{-\infty}^{0} dk_y\left(\frac{e^{ik_y(r_y-r_x)}}{-2k_y+i\varepsilon} + \frac{e^{ik_y(r_y+r_x)}}{2k_y+i\varepsilon}\right)\right.$$

$$\left. + \int_{0}^{\infty} dk_y\left(\frac{e^{ik_y(r_y+r_x)}}{2k_y+i\varepsilon} + \frac{e^{ik_y(r_y-r_x)}}{-2k_y+i\varepsilon}\right)\right\}\theta(r_x)$$

$$= \frac{i}{2\pi}\cdot\frac{1}{2}\left\{\int_{-\infty}^{\infty} dk_y\,\frac{e^{ik_y(r_y+r_x)}}{k_y+i\varepsilon} - \int_{-\infty}^{\infty} dk_y\,\frac{e^{ik_y(r_y-r_x)}}{k_y-i\varepsilon}\right\}\theta(r_x)$$

$$= \left(-\frac{1}{2}\right)\left\{-\theta(-r_y-r_x) - \theta(r_y-r_x)\right\}\theta(r_x)$$

$$= \frac{1}{2}\left\{\theta(-r_y-r_x) + \theta(r_y-r_x)\right\}\theta(r_x) = \frac{1}{2}\,\theta(|r_y|-r_x)\,\theta(r_x)$$



14.(d)

$$\frac{1}{2\pi i}\int_{-\infty}^{\infty} ds\,\frac{e^{its}}{s\pm i\varepsilon} = \mp\theta(\mp t)$$

28

d) $G_s = \frac{1}{(2\pi)^2}\int \frac{e^{i\vec{k}(\vec{r}-\vec{r}')}}{k_x^2+k_y^2}\,d\vec{k} = \frac{1}{(2\pi)^2}\int \frac{e^{ikR\cos\theta}}{k^2}\cdot k\,dk\,d\theta = \frac{1}{2\pi}\int_0^\infty \frac{dk}{k}\underbrace{\int_0^{2\pi} d\theta \frac{e^{ikR\cos\theta}}{2\pi}}_{J_0(kR)}$

$J_0(kR) = 1 + \sum_{n=1}^{\infty} \frac{(-1)^n}{(n!)^2}\left(\frac{kR}{2}\right)^{2n}$

$G_s = G - \frac{1}{2\pi}\int_0^\infty \frac{dk}{k}$

$J_n(z) = \frac{1}{\pi i^n}\int_0^\pi e^{iz\cos\theta}\cos(n\theta)\,d\theta$ (Hansen d'ordre n)

1° $G(R) = G_s(R) + (R\text{-independent})$

$\frac{\partial G(R)}{\partial R} = \frac{\partial G_s(R)}{\partial R}$

"

$\frac{1}{2\pi}\int_0^\infty \frac{1}{k}\underbrace{\frac{\partial}{\partial R}J_0(kR)}_{k J_0'(kR) = \frac{k}{R}\frac{\partial J_0(kR)}{\partial k}}dk$

$= \frac{1}{2\pi}\frac{1}{R}\int_0^\infty \frac{\partial J_0(kR)}{\partial k}\,dk$

$= \frac{1}{2\pi}\cdot\frac{1}{R}\cdot J_0(kR)\Big|_0^\infty = -\frac{1}{2\pi}\frac{1}{R}$

$G_s(R) = -\frac{1}{2\pi}\int\frac{dR}{R} = -\frac{1}{2\pi}\ln R$

$G(R) = -\frac{1}{2\pi}\ln R + (R\text{-independent } \infty)$

2° $G_s(R) = \frac{1}{2\pi}\int_0^\infty \frac{J_0(kR) - J_0(kR_0)}{k}dk = G(R) - G(R_0)$

$= \frac{1}{2\pi}\int_0^\infty \frac{dk}{k}\int_{R_0}^{R} dR \underbrace{\frac{\partial}{\partial R}J_0(kR)}_{k J_0'(kR) = \frac{k}{R}\frac{\partial J_0(kR)}{\partial k}}$

$= \frac{1}{2\pi}\int_{R_0}^{R} dR\,\frac{1}{R}J_0(kR)\Big|_{k=0}^{k=\infty}$

$= -\frac{1}{2\pi}\int_{R_0}^{R}\frac{dR}{R} = -\frac{1}{2\pi}\ln\left(\frac{R}{R_0}\right)$

f) $G_{1s} = -\frac{1}{(2\pi)^3}\int \frac{e^{i\vec{k}(\vec{r}-\vec{r}')}}{|\vec{k}|^2}\,d\vec{k}$

$= -\frac{1}{(2\pi)^3}\int_0^\infty k^2\,dk\int_0^\pi \sin\theta\,d\theta\int_0^{2\pi} d\varphi\,\frac{e^{ikR\cos\theta}}{k^2}$

$= -\frac{1}{(2\pi)^2}\int_0^\infty dk\int_{-1}^{1} d\mu\, e^{ikR\mu}$

$\frac{e^{ikR} - e^{-ikR}}{ikR} = \frac{2}{kR}\sin kR$

$= -\frac{1}{2\pi^2}\cdot\frac{1}{R}\int_0^\infty dk\,\frac{\sin kR}{k} = -\frac{1}{2\pi^2}\frac{1}{R}\underbrace{\int_0^\infty dx\,\frac{\sin x}{x}}_{\frac{\pi}{2}} = -\frac{1}{4\pi}\cdot\frac{1}{R}$

29

a)

$L_{x,y} = D_x^2 - D_y$

(1) $L_{x,y}\, G_1(x,y;x',y') = \delta(x-x')\,\delta(y-y')$

$$G_1(x,y;x',y') = -\theta(y-y')\,\frac{1}{\sqrt{4\pi|y-y'|}}\,e^{-\frac{(x-x')^2}{4|y-y'|}}$$

(2) $\boxed{G_1(0,y;x',y') = 0}$

$L_{xy}\, u(x,y) = f(x,y)$

$u(x,y) = \int dx'dy'\, G_1(x,y;x',y')\, f(x',y')$

$u(0,y) = 0$

$G(x,y;x',y') = \left[G_1(x,y;x',y') - G_1(x,y;-x',y')\right]\theta(x')$

$x' > 0$ and $x > 0$

$L_{xy}\, G(x,y;x',y') = L_{xy}\, G_1(x,y,x',y') - L_{xy}\, G_1(x,y;-x',y')$

$= \delta(x-x')\,\delta(y-y') - \underbrace{\delta(x+x')}_{0}\,\delta(y-y')$

$= \delta(x-x')\,\delta(y-y')$

29. $L = D_x^2 - D_y$ $\qquad G_s(x,y;x',y') = -\frac{1}{\sqrt{4\pi|y-y'|}} e^{-\frac{(x-x')^2}{4|y-y'|}} \cdot \theta(y-y')$

a) $G(x,y;x',y') = \left(G_s(x,y;x',y') - G_s(x,y;-x',y')\right)\theta(x)$

b) $y = k^2 t$, $u(x,t) = u(x,y) = \int_{-\infty}^{\infty} G(x,y;x',y') \left(-f(x')\,\delta(y')\right) dx'dy'$

$$= -\int_0^{\infty} G(x,y;x',y')\, f(x')\,dx'$$

$$= \frac{1}{k\sqrt{4\pi t}} \int_0^{\infty} \left(e^{-\frac{(x-x')^2}{4k^2 t}} - e^{-\frac{(x+x')^2}{4k^2 t}} \right) f(x')\,dx'$$

$L\cdot u = \int_0^{\infty} L\cdot G\, f(x')dx' = f(x)\,\delta(t) = 0 \quad (t>0)$

$u(x,t) = 0$

$u(x,\epsilon) = \int_0^{\infty} \left(G_s(x,\epsilon;x',0) - G_s(x,\epsilon;-x',0) \right) f(x')\,dx'$

$= \int_{-\infty}^{\infty} G_s(x,\epsilon;x',0)\, \tilde{f}(x')\,dx'$ $\qquad \tilde{f}(x) = \begin{cases} f(x) & (x>0) \\ -f(x) & (x<0) \end{cases}$

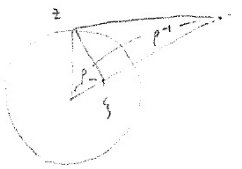
$= \hat{f}(x) = f(x) \quad (x>0)$

c) $G(x,y;x',y') = -\frac{1}{2\pi}\left\{ \ln\sqrt{(x-x')^2+(y-y')^2} - \ln\sqrt{(x-x')^2+(y+y')^2} \right\}$

$(\partial_x^2 + \partial_y^2)\, G = \delta(x-x')\left[\delta(y-y') - \delta(y+y')\right] = \delta(x-x')\,\delta(y-y')$

$\quad\quad\quad

29.



$\bar{\zeta}^{-1} = \omega$

単位円の方程式

$|z-\zeta| = |\zeta|\,|z-\omega|$

$\zeta = \rho e^{i\theta} \quad (\rho < 1)$

円: $|z-\zeta| = k\,|z-\omega|$

kを求める: $(1-\rho)^2 = k^2\left(\frac{1}{\rho}-1\right)^2 \;\to\; k = \rho = |\zeta|$

$G_\zeta(z,\zeta) = \frac{1}{2\pi}\ln\frac{R}{r_0}$

$L_z\, G_\zeta(z,\zeta) = \delta^{(2)}(z-\zeta)$

$G(z,\zeta) = G_\zeta(z,\zeta) - G_\zeta(z,\omega)$

$G(z,\zeta) = G_\zeta(z,\zeta) - G_\zeta(z,\omega) - \frac{1}{2\pi}\ln|\zeta|$

$= \frac{1}{2\pi}\ln|z-\zeta| - \frac{1}{2\pi}\ln|z-\omega| - \frac{1}{2\pi}\ln|\zeta|$

$= \frac{1}{2\pi}\ln\frac{|z-\zeta|}{|\zeta|\,|z-\omega|} \longrightarrow G(|z|=1,\zeta) = 0$

$L_z\, G(z,\zeta) = \delta^{(2)}(z-\zeta) - \delta^{(2)}(z-\omega) = \delta^{(2)}(z-\zeta)$

$|z|<1$

$|\omega|>1$

c) If $\frac{1}{|\vec{r}-\vec{r}'|} = \frac{k}{|\vec{r}'|}\cdot\frac{1}{\left|\vec{r}-\frac{k^2}{|\vec{r}'|^2}\vec{r}'\right|}$ for $|\vec{r}'| = R$ 를 만족하는 k...

$|\vec{r}-\vec{r}'|^2 = r^2 + r'^2 - 2\vec{r}\cdot\vec{r}' = R^2 + r'^2 - 2rr'\cos\theta$

$\frac{|\vec{r}'|^2}{R^2}\left|\vec{r}-\frac{k^2}{|\vec{r}'|^2}\vec{r}'\right|^2 = \frac{r'^2}{R^2}\left(R^2 - 2\frac{R^2}{r'^2}rR\cos\theta + \frac{R^4}{r'^4}r'^2\right) = r'^2 - 2r'R\cos\theta + 1$

$\Delta\frac{1}{|\vec{r}-\vec{r}'|} = -4\pi\,\delta(\vec{r}-\vec{r}')$

Yukawa pot $mc^2\frac{e^{-x}}{x} \quad (x = mr)$

$(\Delta - m^2)\frac{e^{-m|\vec{r}-\vec{r}'|}}{|\vec{r}-\vec{r}'|} = -4\pi\,\delta(\vec{r}-\vec{r}')$

30.

$$\begin{cases} H_r = \frac{1}{r\sin\theta}\frac{\partial}{\partial\theta}(\sin\theta A_\varphi) = -\frac{\partial}{r^2} \\ H_\theta = -\frac{1}{r}\frac{\partial}{\partial r}(rA_\varphi) = 0 \\ H_\varphi = 0 \end{cases}$$

$\rightarrow$ 2つの2" non singular

$$\vec{\partial} = \left(-\frac{\partial}{r(1-z)} + \frac{y}{r(1-z)}, \frac{\partial}{\partial z}\right)$$

$$\theta' \sim \sin\theta = o(x^2+y^2)$$

$$\frac{1}{r(1-z)} = \frac{r+z}{r} \cdot \frac{1}{x^2+y^2} = \frac{1}{x^2+y^2}\{1-o(x^2+y^2)\} + \frac{z}{r(x^2+y^2)}\{1+o(x^2+y^2)\}$$

$$H = \left(-D_y\frac{x}{r(1-z)}, -D_x\frac{y}{r(1-z)}, D_x\frac{x}{r(1-z)} - D_y\frac{y}{r(1-z)}\right)$$

$H_x = H_y = 0 \quad ((x,y)\to(0,0))$ $\qquad H_z = 4\pi\delta(x)\delta(y)$

$$\therefore H = -\frac{\partial}{r^2}e_r + 4\pi y\,\delta(x)\delta(y)\theta(z)e_z$$

直交座標 $\longleftrightarrow$ 極座標

[2次元]

$$\vec{r} = xe_x + ye_y \qquad e_y \rightarrow e_x$$

$$= \begin{pmatrix} x \\ y \end{pmatrix}$$

$$= \begin{pmatrix} r\cos\theta \\ r\sin\theta \end{pmatrix} \qquad e_\theta \rightarrow e_r \rightarrow r$$

$$= Ae_r + Be_\theta$$

$$= (A\cos\theta - B\sin\theta)e_x + (A\sin\theta + B\cos\theta)e_y$$

$$e_r = \begin{pmatrix} \cos\theta \\ \sin\theta \end{pmatrix} = \cos\theta\cdot e_x + \sin\theta\, e_y$$

$$e_\theta = \begin{pmatrix} -\sin\theta \\ \cos\theta \end{pmatrix} = -\sin\theta\, e_x + \cos\theta\, e_y$$

$$\begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}\begin{pmatrix} A \\ B \end{pmatrix}$$

$$A = x\cos\theta + y\sin\theta$$

$$B = -x\sin\theta + y\cos\theta$$

$$\therefore A_r = A_x\cos\theta + A_y\sin\theta$$

$$A_\varphi = -A_x\sin\theta + A_y\cos\theta$$

30

[3次元]

$$\mathbf{e}_r = \begin{pmatrix} \sin\theta\cos\varphi \\ \sin\theta\sin\varphi \\ \cos\theta \end{pmatrix}$$

$\downarrow\ \theta \to \theta + \frac{\pi}{2}$

$$\mathbf{e}_\theta = \begin{pmatrix} \cos\theta\cos\varphi \\ \cos\theta\sin\varphi \\ -\sin\theta \end{pmatrix}$$



$$\mathbf{e}_\varphi = \mathbf{e}_r \times \mathbf{e}_\theta = \begin{pmatrix} -\sin\varphi \\ \cos\varphi \\ 0 \end{pmatrix}$$

$$\frac{1+\cos\theta}{\sin\theta} = \frac{\sin\theta}{1-\cos\theta}$$

$$A = \frac{\gamma}{r}\begin{pmatrix} -\frac{(1+\cos\theta)\sin\varphi}{r\sin^2\theta} \\ \frac{(1+\cos\theta)\cos\varphi}{r\sin^2\theta} \\ 0 \end{pmatrix} = \frac{\gamma}{r}\begin{pmatrix} -\frac{\sin\theta\sin^2\varphi}{1-\cos\theta} \\ \frac{\sin\theta\cos\varphi}{1-\cos\theta} \\ 0 \end{pmatrix} = \gamma\begin{pmatrix} -\frac{y}{r(r-z)} \\ \frac{x}{r(r-z)} \\ 0 \end{pmatrix}$$

$$(D_x^2 + D_y^2)\left(\ln|x^2+y^2|\right) = D_x\left(\frac{2x}{x^2+y^2}\right) + D_y\left(\frac{2y}{x^2+y^2}\right) = 4\pi\,\delta(x)\,\delta(y)$$

[30]

$$A = g\frac{1+\cos\theta}{r\sin\theta}e_\phi \qquad \begin{pmatrix}x\\y\\z\end{pmatrix} = re_r = r\begin{pmatrix}\sin\theta\cos\phi\\ \sin\theta\sin\phi\\ \cos\theta\end{pmatrix} \quad , \quad e_\phi = \begin{pmatrix}-\sin\phi\\ \cos\phi\\ 0\end{pmatrix} = \frac{1}{r\sin\theta}\begin{pmatrix}-y\\ x\\ 0\end{pmatrix}$$

$$= g\frac{1+\frac{z}{r}}{x^2+y^2}\begin{pmatrix}-y\\ x\\ 0\end{pmatrix} = \frac{g}{r(r-z)}\begin{pmatrix}-y\\ x\\ 0\end{pmatrix}$$

$$\nabla\times A = \frac{1}{r\sin\theta}\left[\frac{\partial}{\partial\theta}(\sin\theta\cdot A_\phi)\right]e_r = -\frac{g}{r^2}e_r$$

$$(D_x^2 + D_y^2)\left(\ln|x^2+y^2|\right) = 4\pi\,\delta(x)\,\delta(y)$$

31.

a) $P(z) = (z-a)^{-1} \sum P_n (z-a)^n = \sum P_n (z-a)^{n-1}$

$Q(z) = (z-a)^{-2} \sum Q_n (z-a)^n = \sum Q_n (z-a)^{n-2}$

$w = (z-a)^{\lambda} \sum A_n (z-a)^n$

$$w'' + Pw' + Qw = \sum (n+\lambda)(n+\lambda-1) A_n (z-a)^{n+\lambda-2} + \sum\sum (n+\lambda) A_n (z-a)^{n+\lambda-1} \cdot P_{n'} (z-a)^{n'-1}$$

$$+ \sum\sum A_n (z-a)^{n+\lambda} \cdot Q_{n'} (z-a)^{n'-2}$$

$$= \sum_{n=0}^{\infty} (z-a)^{n+\lambda-2} \left(A_n (\lambda+n)(\lambda+n-1) + \sum_{n'=0}^{n} A_{n-n'} (\lambda+n-n') P_{n'} + \sum_{n'=0}^{\infty} A_{n-n'} Q_{n'} \right)$$

$= 0$

$n=0$: $\lambda(\lambda-1) + \lambda P_0 + Q_0 = 0$ $\rightarrow$ $\lambda^{(1)} + \lambda^{(2)} = 1 - P_0$, $\lambda^{(1)} \lambda^{(2)} = Q_0$ 指数に対する

$$A_n (\lambda+n)(\lambda+n-1) + \sum_{n'=0}^{n} A_{n-n'} \left\{ (\lambda+n-n') P_{n'} + Q_{n'} \right\} = 0$$

$$A_n \left\{ (\lambda+n)(\lambda+n-1) + (\lambda+n) P_0 + Q_0 \right\} = - \sum_{n'=1}^{n} A_{n-n'} \left\{ (\lambda+n-n') P_{n'} + Q_{n'} \right\}$$

$n(n+2\lambda-1+P_0)$

$\begin{cases} n(n+\lambda^{(1)} - \lambda^{(2)}) & (\lambda = \lambda^{(1)}) \\ n(n+\lambda^{(2)} - \lambda^{(1)}) & (\lambda = \lambda^{(2)}) \end{cases}$

$$A_n = - \frac{\sum_{n'=1}^{n} \left\{ (\lambda+n-n') P_{n'} + Q_{n'} \right\} A_{n-n'}}{n(n+2\lambda-1+P_0)}$$

但し $\lambda^{(1)} - \lambda^{(2)} \neq$ 整数

31. b) $z = \frac{1}{\zeta}$

$$\frac{d}{dz} = -\zeta^2 \frac{d}{d\zeta} \qquad \frac{d^2}{dz^2} = 2\zeta^3 \frac{d}{d\zeta} + \zeta^4 \frac{d^2}{d\zeta^2} = \zeta^2 \frac{d}{d\zeta}\left(-\zeta^2 \frac{d}{d\zeta}\right)$$

$$\omega'' + P\omega' + Q = 2\zeta^3 \frac{d\omega}{d\zeta} + \zeta^4 \frac{d^2\omega}{d\zeta^2} - P\left(\frac{1}{\zeta}\right)\zeta^2 \frac{d\omega}{d\zeta} + Q\left(\frac{1}{\zeta}\right)\omega = 0$$

$$\therefore \quad \frac{d^2\omega}{d\zeta^2} + \left(\frac{2}{\zeta} - \frac{1}{\zeta^2} P\left(\frac{1}{\zeta}\right)\right)\frac{d\omega}{d\zeta} + \frac{1}{\zeta^4} Q\left(\frac{1}{\zeta}\right)\omega = 0$$

$$\left[\ \frac{1}{\zeta} P\left(\frac{1}{\zeta}\right) \to \text{finite} \quad , \quad \frac{1}{\zeta^2} Q\left(\frac{1}{\zeta}\right) \to \text{finite}\ \right]$$

$$\Rightarrow \left[\ z\,P(z) \to \text{finite} \qquad z^2 Q(z) \to \text{finite}\ \right]$$

$$P(z) = \frac{1}{z}\sum_{n=0}^{\infty} P_n^{(\infty)} \left(\frac{1}{z}\right)^n \qquad Q(z) = \frac{1}{z^2}\sum_{n=0}^{\infty} Q_n^{(\infty)} \left(\frac{1}{z}\right)^n \qquad \omega = z^{-\lambda}\sum A_n^{(\infty)} \left(\frac{1}{z}\right)^n \quad (\lambda > 0)$$

$z = \infty$ が正則点でない

すなわち $P(z) = \frac{2}{z} + \cdots$

$Q(z) = \frac{1}{z^2} + \cdots$

$$\lambda(\lambda+1) - \lambda P_0^{(\infty)} + Q_0^{(\infty)} = 0$$

$$A_n = \frac{\sum_{n'=1}^{n} \left\{(\lambda + n - n')P_{n'}^{(\infty)} - Q_{n'}^{(\infty)}\right\} A_{n-n'}^{(\infty)}}{n(n+2\lambda+1) - P_0^{(\infty)}}$$

c) $\Psi(z) = \prod_{i=1}^{m}(z - a_i) \sim z^m$

$\left.\begin{matrix}\Psi(z)P(z) \\ \Psi^2(z)Q(z)\end{matrix}\right\}$ が1価正則 $\to$ $\Psi(z)P(z) = \sum_{n=0}^{\infty} P_n z^n$, $\Psi(z)Q(z) = \sum_{n=0}^{\infty} q_n z^n$

$$P(z) = \frac{\sum_{n=0}^{\infty} P_n z^n}{\Psi(z)} = \frac{1}{\Psi(z)} \sum_{n=0}^{m-1} P_n z^n$$

$$Q(z) = \frac{\sum_{n=0}^{\infty} q_n z^n}{\Psi^2(z)} = \frac{1}{\Psi(z)^2} \sum_{n=0}^{2m-2} q_n z^n$$

☺ $z = \infty$ が確定特異点

$$P(z) = \frac{\sum_{n=0}^{m-1} P_n z^n}{\prod_{i=1}^{m}(z - a_i)} = \sum_{k=1}^{m} \frac{A_k}{z - a_k}$$

$$Q(z) = \frac{\sum_{n=0}^{2m-2} q_n z^n}{\prod_{i=1}^{m}(z - a_i)^2} = \sum_{k=1}^{m} \left\{ \frac{C_k}{z - a_k} + \frac{B_k}{(z - a_k)^2} \right\}$$

[Fucks型 微分方程式]

有限個の確定特異点のみを特異点としてもつ線形方程式 ($z=\infty$ を含めていう)

$m=1$ の場合 $\boxed{z=a,\ \infty}$ が確定特異点

$$\frac{d^2w}{dz^2}+\frac{A}{z-a}\frac{dw}{dz}+\frac{B}{(z-a)^2}w=0$$

指数

$z=a$ $\begin{pmatrix}\lambda\\ \mu\end{pmatrix}$

$z=\infty$ $\begin{pmatrix}\lambda_\infty\\ \mu_\infty\end{pmatrix}$

$\lambda+\mu=1-A$ $\quad\lambda_\infty+\mu_\infty=A-1$

$\lambda\mu=B$ $\quad\lambda_\infty\mu_\infty=B$ $\quad\rightarrow\begin{pmatrix}\lambda_\infty=-\lambda\\ \mu_\infty=-\mu\end{pmatrix}$ とおきなおせる

$$\boxed{\frac{d^2w}{dz^2}+\frac{1-\lambda-\mu}{z-a}\frac{dw}{dz}+\frac{\lambda\mu}{(z-a)^2}w=0}$$

f)

$\begin{pmatrix}a\to 0\\ \infty\to 1\end{pmatrix}$ $\quad\xi=\dfrac{z-a}{z}$ $\quad z=\begin{pmatrix}a\\ \infty\end{pmatrix}\to\xi=\begin{pmatrix}0\\ 1\end{pmatrix}\longrightarrow\xi=\begin{pmatrix}a_1\\ a_2\end{pmatrix}$ $\quad\dfrac{\xi-a_1}{a_2-a_1}=\xi$

$m=2$ の場合 $z=a_1,a_2,\infty$ が確定特異点

$$\frac{d^2w}{dz^2}+\left(\frac{A_1}{z-a_1}+\frac{A_2}{z-a_2}\right)\frac{dw}{dz}+\left(\frac{C_1}{z-a_1}+\frac{B_1}{(z-a_1)^2}+\frac{C_2}{z-a_2}+\frac{B_2}{(z-a_2)^2}\right)w=0$$

$$\frac{C_1}{z-a_1}+\frac{C_2}{z-a_2}=\frac{\overset{0}{(C_1+C_2)}z\ \overbrace{-C_1a_2-C_2a_1}^{B_3}}{(z-a_1)(z-a_2)}$$

$$\frac{d^2w}{dz^2}+\left(\frac{A_1}{z-a_1}+\frac{A_2}{z-a_2}\right)\frac{dw}{dz}+\left(\frac{B_1}{(z-a_1)^2}+\frac{B_2}{(z-a_2)^2}+\frac{B_3}{(z-a_1)(z-a_2)}\right)w=0$$

$\lambda_1+\mu_1+\lambda_2+\mu_2+\lambda_\infty+\mu_\infty=2-1=1$

$A_1=1-\lambda_1-\mu_1$ $\quad A_2=1-\lambda_2-\mu_2$

$B_1=\lambda_1\mu_1$ $\quad B_2=\lambda_2\mu_2$

$\lambda_\infty\mu_\infty=B_1+B_2+B_3$

($z=a_1,a_2$ 確定特異点 / $z=\infty$ 正則点)

$$\rightarrow\begin{cases}A_1+A_2=2\\ B_1+B_2+B_3=0\\ a_1B_1+a_2B_2+(a_1+a_2)B_3=0\end{cases}$$ を示せ

$z=a_1$ $\begin{pmatrix}\lambda\\ \mu\end{pmatrix}$ とすると $z=a_2$ $\begin{pmatrix}-\lambda\\ -\mu\end{pmatrix}$ を示せ

$z=\infty$ (正則点) の指数は $\begin{pmatrix}1\\ 0\end{pmatrix}$

31 d)　　m+1個の確定特異点

とする $\begin{cases} a_1, a_2, \cdots, a_m \quad (a_1 \neq a_2 \neq \cdots \neq a_m) & \text{確定特異点} \\ z=\infty \text{ が確定特異点} \end{cases}$

$$\begin{cases} \dfrac{d^2w}{dz^2} + \sum_{k=1}^{m} \dfrac{A_k}{z-a_k}\dfrac{dw}{dz} + \sum_{k=1}^{m}\left[\dfrac{C_k}{z-a_k} + \dfrac{B_k}{(z-a_k)^2}\right]w = 0 \\ \sum_{k=1}^{m} C_k = 0 \end{cases}$$

指数の関係式

$$\begin{cases} \lambda_k + \mu_k = 1 - A_k \\ \lambda_k \mu_k = B_k \end{cases} \quad k=1\sim m$$

$$\begin{cases} \lambda_\infty + \mu_\infty = \sum_{k=1}^{m} A_k - 1 \\ \lambda_\infty \mu_\infty = \sum_{k=1}^{m} (a_k C_k + \beta_k) \end{cases}$$

$$\sum_{k=1}^{m} \left[\frac{1}{z}\cdot\frac{C_k}{1-\frac{a_k}{z}} + \frac{1}{z^2}\cdot\frac{B_k}{\left(1-\frac{a_k}{z}\right)^2}\right]$$

$$= \sum_{k=1}^{\infty}\left[\frac{1}{z}\left(1+\frac{a_k}{z}+\left(\frac{a_k}{z}\right)^2+\cdots\right)C_k + \frac{1}{z^2}\left(1+2\frac{a_k}{z}+\cdots\right)B_k\right]$$

（$\frac{1}{z}$ の項は $\sum C_k = 0$ より 0）

$$= \frac{1}{z^2}\sum_{k=1}^{m}(a_k C_k + B_k) + \frac{1}{z^3}\sum_{k=1}^{m}(a_k^2 C_k + 2a_k B_k) + \cdots$$

$$\boxed{\sum_{k=1}^{m}(\lambda_k + \mu_k) + (\lambda_\infty + \mu_\infty) = m-1} \qquad \text{Fuchs の関係}$$

$$\begin{cases} A_k = 1 - \lambda_k - \mu_k \\ B_k = \lambda_k \mu_k \end{cases}$$

$z=\infty$ が正則点である

$z = \frac{1}{\zeta}$

$\dfrac{2}{\zeta} - \dfrac{1}{\zeta^2}P\left(\dfrac{1}{\zeta}\right)$
$\dfrac{1}{\zeta^4}Q\left(\dfrac{1}{\zeta}\right)$
} が $\zeta=0$ で正則 $\iff$ $\begin{cases} P(z) = \dfrac{2}{z} + O\left(\dfrac{1}{z^2}\right) \\ Q(z) = O\left(\dfrac{1}{z^4}\right) \end{cases}$

$z=\infty$ が正則点である

$$\begin{cases} \sum_{k=1}^{m} A_k = 2 \\ \sum_{k=1}^{m}(a_k C_k + \beta_k) = 0 \qquad \left(\sum_{k=1}^{m} C_k = 0\right) \\ \sum_{k=1}^{m}(a_k^2 C_k + 2a_k B_k) = 0 \end{cases}$$

31 e) $z=\infty$ の正則性　　$z=\frac{1}{\zeta}$

$$\frac{d^2\omega}{d\zeta^2}+\underbrace{\left(\frac{2}{\zeta}-\frac{1}{\zeta^2}P\left(\frac{1}{\zeta}\right)\right)}_{\tilde{P}(\zeta)}\frac{d\omega}{d\zeta}+\underbrace{\frac{1}{\zeta^4}Q\left(\frac{1}{\zeta}\right)}_{\tilde{Q}(\zeta)}\omega=0$$

$\zeta=0$ の正則性 $\Rightarrow$ $\tilde{P}(0)$, $\tilde{Q}(0)$ が finite

$P\left(\frac{1}{\zeta}\right)=2\zeta+\cdots$

$Q\left(\frac{1}{\zeta}\right)=\Delta\zeta^5+\cdots$

$z=a$ が唯一の特異点 $\to$ $P(z)=\frac{1}{z-a}\sum_{n=0}^{\infty}P_n(z-a)^n=\frac{2}{z-a}$

$Q(z)=0$

$$\frac{d^2\omega}{dz^2}+\frac{2}{z-a}\frac{d\omega}{dz}=0$$

indicial equation　$\lambda(\lambda-1)+2\lambda=0$ $\to$ $\lambda=0,-1$

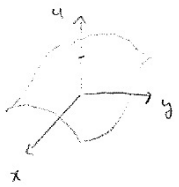
bilinear 変換　$x=\frac{1}{z-a}$　　$\frac{d^2\omega}{dx^2}=0$ $\to$ $\omega=C_1x+C_2$

$\therefore \omega=\frac{C_1}{z-a}+C_2$

$$\begin{cases} A_1=2 \\ \alpha_1C_1+\beta_1=0 \\ \alpha_1^2C_1+2\alpha_1\beta_1=0 \end{cases}
\begin{cases} C_1=0 \\ \beta_1=0 \\ A_1=2 \end{cases}$$

$$\frac{d^2\omega}{dz^2}+\frac{2}{z-a}\frac{d}{dz}\omega=0$$

[1] $A u_x + B u_y = C$



$u = u(x,y)$: 曲面の方程式

$du = u_x dx + u_y dy$

$u - u_0 = u_x (x - x_0) + u_y (y - y_0)$ $(x_0\ y_0\ u_0)$ での接面の方程式

$u_x \Delta x + u_y \Delta y + (-1)\Delta u = 0$

$\begin{pmatrix} \Delta x \\ \Delta y \\ \Delta u \end{pmatrix} \perp \begin{pmatrix} u_x \\ u_y \\ -1 \end{pmatrix}$ ＝接線ベクトルに直交

$A u_x + B u_y + C(-1) = 0$

$\begin{pmatrix} A \\ B \\ C \end{pmatrix} \perp \begin{pmatrix} u_x \\ u_y \\ -1 \end{pmatrix}$ ベクトル場 $\begin{pmatrix} A(x,y,u) \\ B \\ C \end{pmatrix}$ モンジュ Monge の方向場

～接平面内にある

[特性曲線]

$\begin{cases} \dfrac{dx(t)}{dt} = A(x(t), y(t), u(t)) \\ \dfrac{dy}{dt} = B \\ \dfrac{du}{dt} = C \end{cases}$

(1) 特性曲線が解曲面の中にある事

証明 $u = u(x,y)$ を解曲面とする

$$\frac{d}{dt}\{u(t) - u(x(t), y(t))\} = \frac{du(t)}{dt} - u_x(x(t)\,y(t))\frac{dx(t)}{dt} - u_y(x(t)\,y(t))\frac{dy(t)}{dt}$$

$$= C(x(t)\,y(t)\,u(t)) - u_x(x(t)\,y(t))A(x(t)\,y(t)\,u(t)) - u_y(x(t)\,y(t)\,u(t))B(x(t)\,y(t)\,u(t))$$

$$= 0$$

$u(t) = u(x(t), y(t)) = \text{const}$

$t=0$ で決定

$u_0 = u(x_0\ y_0) + \text{const} \rightarrow \text{const} = 0 \Rightarrow u(t) = u(x(t)\,y(t))$

$x(s,0) = x_0(s)$
$y(s,0) = y_0(s)$
$u(s,0) = u_0(s)$

$(x_0(s)\ y_0(s)\ u_0(s))$
初期曲線

(2) s を連続的に変えると曲線 C を通る特性曲線の全体は一般に曲面を構成する ：これを解曲面という

証明 $\begin{cases} x = x(s,t) \\ y = y(s,t) \end{cases}$ が s,t について逆に解ける（仮定する） $\begin{pmatrix} s = s(x,y) \\ t = t(x,y) \end{pmatrix}$ ← Jacobian $\begin{vmatrix} \frac{\partial x(s,t)}{\partial s} & \frac{\partial x}{\partial t} \\ \frac{\partial y}{\partial s} & \frac{\partial y}{\partial t} \end{vmatrix} \neq 0$

$u = u(s(x,y), t(x,y)) = f(x,y)$

$\dfrac{\partial}{\partial t} f(x(s,t), y(s,t)) = f_x(x(s,t), y(s,t))\dfrac{\partial x(s,t)}{\partial t} + f_y(\quad)\dfrac{\partial y(s,t)}{\partial t}$

$= A f_x + B f_y$

$x_s \frac{\partial y}{\partial t} - \frac{\partial x}{\partial t} y_s \neq 0$

特性曲線を x-y ... ← 基本特性曲線

$x_s B - y_s A \neq 0$ $x_s : y_s \neq A : B$

特性曲線を x-y 平面に射影した時の接線の方向

[2]

(a) $au_x + bu_y = c$ $\frac{du}{dt} = a$, $\frac{dy}{dt} = b$, $\frac{du}{dt} = c$

$x = x_0 + at$
$y = y_0 + bt$
$u = u_0 + ct \rightarrow t = \frac{1}{c}(u - u_0)$

$x = (x_0 - \frac{a}{c}u_0) + \frac{a}{c}u = \alpha + \frac{a}{c}u$

$y = (y_0 - \frac{b}{c}u_0) + \frac{b}{c}u = \beta + \frac{b}{c}u$

$\beta = F(\alpha)$ $x - \frac{a}{c}u = F(y - \frac{b}{c}u)$

すなわち

$F(\alpha, \beta) = 0$

$F(x - \frac{a}{c}u, y - \frac{b}{c}u) = 0$

x, y で偏微分

$x: F_\alpha (1 - \frac{a}{c}u_x) + F_\beta (-\frac{b}{c}u_y) = 0$

$y: F_\alpha (-\frac{a}{c}u_y) + F_\beta (1 - \frac{b}{c}u_y) = 0$

$$\begin{pmatrix} 1 - \frac{a}{c}u_x & -\frac{b}{c}u_y \\ -\frac{a}{c}u_x & 1 - \frac{b}{c}u_y \end{pmatrix} \begin{pmatrix} F_\alpha \\ F_\beta \end{pmatrix} = 0$$

$\Leftrightarrow au_x + bu_y = c$

$b\,dx = a\,dy$

$bx - ay = c_1$

$au - cx = c_2$

$au - cx = \varphi(bx - ay)$

$u = \frac{1}{a}\varphi(bx - ay) + cx$

(b) $u = 0$, $x^2 + y^2 = 1$

$(x - \frac{a}{c}u)^2 + (y - \frac{b}{c}u)^2 = 1$

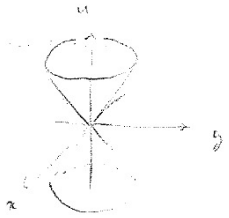
[3] $u = x u_x + y u_y$

$\frac{dx}{dt} = x \quad \frac{dy}{dt} = y \quad \frac{du}{dt} = u$

$x = x_0 e^t \quad y = y_0 e^t \quad u = u_0 e^t$

$u = \frac{u_0}{x_0} x \quad , \quad u = \frac{u_0}{y_0} y$

$\left(\frac{x}{u}\right)^2 + \left(\frac{y}{u}\right)^2 = 1$



[4] $x u_y - y u_x = 0$

$\frac{dx}{dt} = -y \quad , \quad \frac{dy}{dt} = x \quad \frac{du}{dt} = 0$

$x = C_1 \cos(t+\alpha) \quad y = C_1 \sin(t+\alpha) \quad u = C_2$

$u = f(x^2+y^2)$

$\begin{cases} x^2+y^2 = \text{const} = \beta \\ u = \text{const} = \alpha \end{cases} \qquad \alpha = F(\beta)$

$(y=0 \,,\, u_x = 1)$ $\quad t=0 \,;\quad x = s\cos t \quad y = s\sin t \quad u = \frac{1}{s}$

$\Rightarrow x^2 + y^2 = s^2 \qquad u^2 = \frac{1}{x^2+y^2}$

[5] $u_t + uu_x = 0 \longrightarrow u_y + uu_x = 0$

$\frac{dx}{dt} = u \quad \frac{dy}{dt} = 1 \quad \frac{du}{dt} = 0$

$x = x_0 + u_0 t \quad y = y_0 + t \quad u = u_0$

$t=0 \quad u = \bar{U}(x)$

とおける時には $\begin{cases} x_0 = s & \text{とおけばよい} \\ y_0 = 0 \\ u_0 = \bar{U}(s) \end{cases}$

$\begin{cases} x = s + \bar{U}(s)t = x(s,t) \\ y = t = y(s,t) \\ u = \bar{U}(s) = u(s,t) \end{cases}$

$x = s + \bar{U}(s)t$

$y = t$

t を逆解きする

$\Delta(s,t) = x_s y_t - x_t y_s$

$= (1 + \bar{U}'(s)t) \cdot 1 - U(s) \cdot 0$

$= 1 + \bar{U}'(s)t \neq 0$ なら OK

解けるとして

$\begin{cases} s = x - U(s)t \\ t = y \end{cases}$

$u = \bar{U}(s) = \bar{U}(x - uy)$

$u = \bar{U}(x - uy)$ が解曲面

解を支配する情報

$u_y = \bar{U}'(x-uy)(-u_y y - u)$

$= \frac{-u\bar{U}'(x-uy)}{1 + \bar{U}'(x-uy)y}$

$u_x = \bar{U}'(x-uy)(1 - u_x y)$

$= \frac{\bar{U}'(x-uy)}{1 + \bar{U}'(x-uy)y}$ より OK.

普遍特性曲線の包絡線上で

$1 + \bar{U}'(x-uy)y = 0$ である点を

$\begin{cases} x = s + U(s)t \\ y = t \end{cases}$

$x = s + U(s)y$ を決めると決まる (x-y 平面での曲線)

$f(x,y,s) = x - s - U(s)y$

とおくと包絡線は

$f(x,y,s) = 0$

$f_s(x,y,s) = 0$

[7] $u_x + u_y = u^2$

$\frac{dx}{dt} = 1 \quad \frac{dy}{dt} = 1 \quad \frac{du}{dt} = u^2$

$x = x_0 + t \quad y = y_0 + t \quad u = \frac{-1}{t - t_0}$

$\int \frac{du}{u^2} = \int u^{-2} du \quad -\frac{1}{u} = t + A$

$= \frac{1}{-t + \frac{1}{u_0}} \quad t - t_0 = -\frac{1}{u}$

$t = t_0 - \frac{1}{u}$

$x = x_0 + t_0 - \frac{1}{u}$

$y = y_0 + t_0 - \frac{1}{u}$

$x = \alpha - \frac{1}{u}$ $\quad x + \frac{1}{u} = \alpha \quad , \quad y + \frac{1}{u} = \beta$

$y = \beta - \frac{1}{u}$ $\quad \beta = F(\alpha)$ $\quad y + \frac{1}{u} = F\left(x + \frac{1}{u}\right)$

$F\left(x + \frac{1}{u}, y + \frac{1}{u}\right) = 0$

$x = s, \quad y = -s, \quad u = s$

$-s + \frac{1}{s} = F\left(s + \frac{1}{s}\right) \quad F\left(s + \frac{1}{s}, -s + \frac{1}{s}\right) = 0$

$F(2\sinh\alpha, 2\cosh\alpha) = 0$

$s = e^{\alpha}$

$2\sinh\alpha = F(2\cosh\alpha) \qquad F\left(s + \frac{1}{s}\right) = -s + \frac{1}{s} = \sqrt{\left(s + \frac{1}{s}\right)^2 - 2}$

$\cosh^2\alpha - \sinh^2\alpha = 1$

$\begin{cases} x = s + t \\ y = -s + t \\ u = \frac{1}{-t + \frac{1}{s}} \end{cases} \qquad \begin{aligned} s &= \frac{1}{2}(x - y) \\ t &= \frac{1}{2}(x + y) \end{aligned}$

$u = \frac{1}{-\frac{x+y}{2} + \frac{2}{x-y}} = \frac{2(x-y)}{-x^2 + y^2 + 4}$

[8] $uu_x + u_y = 1$

$\frac{dx}{dt} = u \quad \frac{dy}{dt} = 1 \quad \frac{du}{dt} = 1$

$x = x_0 + u_0 t + \frac{1}{2}t^2 \qquad u = u_0 + t$

$y = y_0 + t$

$\begin{cases} x_0 = s^2 \\ y_0 = 2s \\ u_0 = s \end{cases} \quad \begin{cases} x = s^2 + st + \frac{1}{2}t^2 \\ y = 2s + t \\ u = s + t \end{cases} \longrightarrow \begin{cases} s = \frac{y}{2} \pm \sqrt{x - \frac{y^2}{4}} \\ t = \mp\sqrt{4x - y^2} \end{cases}$

$x = s^2 + s(y-2s) + \frac{1}{2}(y-2s)^2$

$2x = 2s^2 + 2sy - 4s^2 + y^2 - 4sy + 4s^2$

$2x = 2s^2 + y^2 - 2sy$

$2s^2 - 2y \cdot s + y^2 - 2x = 0$

$s = \frac{2y \pm \sqrt{4y^2 - 8(y^2 - 2x)}}{4}$

$= \frac{2y \pm \sqrt{16x - 4y^2}}{4}$

$= \frac{y}{2} \pm \sqrt{x - \frac{y^2}{4}}$

$u = \frac{y}{2} \pm \sqrt{x - \frac{y^2}{4}} \mp \sqrt{4x - y^2}$

$= \frac{y}{2} \mp \sqrt{x - \frac{y^2}{4}}$

$t = y - 2s$

$= y - 2\left(\frac{y}{2} \pm \sqrt{x - \frac{y^2}{4}}\right)$

$= \mp\sqrt{4x - y^2}$

$u_x = \mp\frac{1}{2}\left(x - \frac{y}{4}\right)^{-\frac{1}{2}}$

$u_y = \frac{1}{2} \mp \frac{1}{2}\left(x - \frac{y}{4}\right)^{-\frac{1}{2}} \cdot \left(-\frac{y}{2}\right)$

$= \frac{1}{2} \pm \frac{y}{8}\left(x - \frac{y}{4}\right)^{-\frac{1}{2}}$

$uu_x = \left(\frac{y}{2} \mp \sqrt{x - \frac{y^2}{8}}\right)\left(\mp\frac{1}{2}\left(x - \frac{y}{4}\right)^{-\frac{1}{2}}\right)$

$= \mp\frac{y}{4}\left(x - \frac{y}{4}\right)^{-\frac{1}{2}} + \frac{1}{2}$

$\sqrt{x - \frac{y^2}{4}} = 0$ 이 경계 곡선

$s^2 = \frac{(4x)^2}{4}$

특성 방정식의 음함수

$f(x, y, s) = s^2 - ys + \frac{y^2}{2} - x$

$f_s(x, y, s) = 2s - y$

s에 대한 편미분 후 (x, u, y)의 음함수

1/28

2/18

3/4

3/18

0-1 $\sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$

(i) $\sigma_i^\dagger = {}^t(\overline{\sigma_i}) = \sigma_i$

$\sigma_i^\dagger \sigma_i = 1$

$\mathrm{Tr}\,\sigma_i = 0$

$M^\dagger = {}^t M^*$

(ii) $[\sigma_i, \sigma_j] \equiv \sigma_i\sigma_j - \sigma_j\sigma_i = 2i\,\epsilon_{ijk}\sigma_k$

(iii) は自明

(iii) $\sigma_i\sigma_j = \delta_{ij} + i\,\epsilon_{ijk}\sigma_k$

$i=j$ のとき $\sigma_i^2 = 1$

$\sigma_1\sigma_2 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} = \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix} = i\sigma_3$ $\quad \sigma_2\sigma_1 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} = \begin{pmatrix} -i & 0 \\ 0 & i \end{pmatrix} = -i\sigma_3$

$$
\begin{aligned}
n_i \sigma_i \, \underline{\underline{\sigma_j}} \, n_k \sigma_k &= n_i n_k (\delta_{ij} + i \epsilon_{ijl} \sigma_l) \sigma_k \\
&= n_j n_k \sigma_k + i n_i n_k \epsilon_{ijl} \sigma_l \sigma_k \\
&= (n \cdot \sigma) n_j + i n_i n_k \epsilon_{ijl} (\delta_{lk} + i \epsilon_{lkm} \sigma_m) \\
&= (n \cdot \sigma) n_j + i n_i n_k \epsilon_{ijk} - n_i n_k \epsilon_{ijl} \epsilon_{lkm} \sigma_m \\
&= (n \cdot \sigma) n_j + i n_i n_k \epsilon_{ijk} - n_i n_k (\delta_{ik} \delta_{jm} - \delta_{im} \delta_{jk}) \sigma_m \\
&= (n \cdot \sigma) n_j + \underbrace{i n_i n_k \epsilon_{ijk}}_{0} - n_k n_k \sigma_j + n_i n_j \sigma_i \\
&= 2 (n \cdot \sigma) n_j - \sigma_j
\end{aligned}
$$

$$
\begin{pmatrix} e^{i\frac{\gamma}{2}} & \\ & e^{-i\frac{\gamma}{2}} \end{pmatrix} \begin{pmatrix} e^{i\frac{\delta}{2}} \cos\frac{\beta}{2} & e^{-i\frac{\alpha}{2}} \sin\frac{\beta}{2} \\ -e^{i\frac{\delta}{2}} \sin\frac{\beta}{2} & e^{-i\frac{\delta}{2}} \cos\frac{\beta}{2} \end{pmatrix} = \begin{pmatrix} e^{i\left(\frac{\gamma}{2}+\frac{\delta}{2}\right)} \cos\frac{\beta}{2} & e^{i\left(\frac{\gamma}{2}-\frac{\alpha}{2}\right)} \sin\frac{\beta}{2} \\ & \end{pmatrix} \quad \text{o.k.}
$$

0-2 $\sigma \cdot a = a_k \sigma_k = \begin{pmatrix} a_3 & a_1 - i a_2 \\ a_1 + i a_2 & a_3 \end{pmatrix}$

$$(\sigma \cdot a)(\sigma \cdot b) = \sigma_i a_i \sigma_j b_j = (\delta_{ij} + i \epsilon_{ijk} \sigma_k) a_i b_j$$

$$= a \cdot b + i \sigma \cdot (a \times b)$$

(i) $\exp\left(i \frac{\theta}{2} n \cdot \sigma\right) = \left[1 - \frac{(n \cdot \sigma)^2}{2!}\left(\frac{\theta}{2}\right)^2 + \frac{(n \cdot \sigma)^4}{4!}\left(\frac{\theta}{2}\right)^4 \cdots\right] + i\left[(n \cdot \sigma)\frac{\theta}{2} - \frac{(n \cdot \sigma)^3}{3!}\left(\frac{\theta}{2}\right)^3 + \cdots\right]$

$= \cos\frac{\theta}{2} + i (n \cdot \sigma) \sin\frac{\theta}{2}$ ∵ $|n| = 1$

(ii) $\exp\left(i \frac{\theta}{2} n \cdot \sigma\right) \sigma \exp\left(-i \frac{\theta}{2} n \cdot \sigma\right) = \left(\cos\frac{\theta}{2} + i (n \cdot \sigma) \sin\frac{\theta}{2}\right) \sigma \left(\cos\frac{\theta}{2} - i (n \cdot \sigma) \sin\frac{\theta}{2}\right)$

$$= \sigma \cos^2\frac{\theta}{2} + i \sin\frac{\theta}{2} \cos\frac{\theta}{2} \underbrace{\left\{(n \cdot \sigma)\sigma - \sigma(n \cdot \sigma)\right\}}_{\text{II}} + \underbrace{(n \cdot \sigma)\sigma(n \cdot \sigma)}_{\text{II}} \sin^2\frac{\theta}{2}$$

$n_i \sigma_i \sigma_j - \sigma_j n_i \sigma_i$ $\qquad n_i \sigma_i \sigma_j n_k \sigma_k = 2(n \cdot \sigma) n - \sigma$

$= n_i [\sigma_i, \sigma_j]$

$= 2i \epsilon_{ijk} n_i \sigma_k$

$= 2i (\sigma \times n)$

(iii) $e^{i\frac{\gamma}{2}\sigma_3} e^{i\frac{\beta}{2}\sigma_2} e^{i\frac{\alpha}{2}\sigma_3} = \left[\left\{1 - \frac{\sigma_3^2}{2!}\left(\frac{\gamma}{2}\right) + \frac{\sigma_3^4}{4!}\left(\frac{\gamma}{2}\right)^4 \cdots\right\} + i\sigma_3\left(\frac{\gamma}{2} - \frac{\sigma_3^2}{3!}\left(\frac{\gamma}{2}\right)^3 + \cdots\right)\right][\beta][\alpha]$

$$= \left(\cos\frac{\gamma}{2} + i\sigma_3 \sin\frac{\gamma}{2}\right)\left(\cos\frac{\beta}{2} + i\sigma_2 \sin\frac{\beta}{2}\right)\left(\cos\frac{\alpha}{2} + i\sigma_3 \sin\frac{\alpha}{2}\right)$$

$$= \begin{pmatrix} \cos\frac{\gamma}{2} + i\sin\frac{\gamma}{2} & 0 \\ 0 & \cos\frac{\gamma}{2} - i\sin\frac{\gamma}{2} \end{pmatrix}\begin{pmatrix} \cos\frac{\beta}{2} & +\sin\frac{\beta}{2} \\ -\sin\frac{\beta}{2} & \cos\frac{\beta}{2} \end{pmatrix}\begin{pmatrix} \cos\frac{\alpha}{2} + i\sin\frac{\alpha}{2} & 0 \\ 0 & \cos\frac{\alpha}{2} - i\sin\frac{\alpha}{2} \end{pmatrix}$$

$$= \begin{pmatrix} e^{i\frac{\gamma}{2}} & 0 \\ 0 & e^{-i\frac{\gamma}{2}} \end{pmatrix}\begin{pmatrix} \cos\frac{\beta}{2} & \sin\frac{\beta}{2} \\ -\sin\frac{\beta}{2} & \cos\frac{\beta}{2} \end{pmatrix}\begin{pmatrix} e^{i\frac{\alpha}{2}} & 0 \\ 0 & e^{-i\frac{\alpha}{2}} \end{pmatrix} \quad //$$

0-3 $\det \exp(h) = \exp(\mathrm{Tr}\, h)$ $\quad UU^\dagger = 1$

$U^\dagger h U = \begin{pmatrix} \ddots & & \\ & \lambda_i & \\ & & \ddots \end{pmatrix} = \tilde{h}$

$$U^\dagger \exp(h) U = U^\dagger \left(1 + h + \frac{1}{2!}h^2 + \cdots\right) U$$

$$= 1 + U^\dagger h U + \frac{1}{2!}(U^\dagger h U)(U^\dagger h U) + \cdots$$

$$\det \exp(h) = \det\left(1 + \tilde{h} + \frac{1}{2!}\tilde{h}^2 + \cdots\right)$$

$$= \det \begin{pmatrix} \ddots & & \\ & e^{\lambda_i} & \\ & & \ddots \end{pmatrix} = \prod_i e^{\lambda_i} = \exp(\mathrm{Tr}\, h)$$

0-4 $A_{n\times n}$ $B = m\times m$

(1) $(A^1 \otimes B^1)(A^2 \otimes B^2)_{ij} =$

1-1

	単位元	逆元	位数
(1)	1	$(\pm 1)^{-1} = \pm 1$	2
(2)	1	$z^{-1} = z^*$	n
(5)	$\begin{pmatrix} 1 & 2 & \cdots & n \\ 1 & 2 & \cdots & n \end{pmatrix}$	$\begin{pmatrix} a_1 & a_2 & \cdots \\ b_1 & b_2 & \cdots \end{pmatrix}^{-1} = \begin{pmatrix} b_1 & b_2 & \cdots \\ a_1 & a_2 & \cdots \end{pmatrix}$	$n!$

半群 (semi-group)

(3), (4) $\in$ (7)

(他) $\det M = 1$ より 1° 2° 3° 共OK.

$\forall g, h \in G \Rightarrow g \circ h \in G$ を示せばよい

(6) $A, B \in O(n) \Rightarrow {}^t(AB)AB = {}^tB\,{}^tAAB = 1 \quad \therefore AB \in O(n)$

${}^tAA = 1$ は $1+2+\cdots+N$ 個の独立な方程式を出すから $\dim O(n) = \frac{n(n-1)}{2}$

(7) $\det(AB) = 1$ (6) $\in$ (5)

(5)

1-3

H が G の subgroup $\Leftrightarrow \forall a,b \in H,\ ab^{-1} \in H$

$\Leftarrow$ $\exists a \in H$ $aa^{-1} = e \in H$ $\quad \forall a,b,c \in H$ $(ab)c = a(bc)$

$\forall b \in H$ $eb^{-1} = b^{-1} \in H$ $\quad \forall b \in H$ $eb = be = b$

$\therefore \forall a,b \in H$ $a \cdot b^{-1} \in H$ $\quad aa^{-1} = a^{-1}a = e$

$a(b^{-1})^{-1} = ab \in H$

$\Rightarrow$ H の単位元 e_H

$e_H e_H = e_H$ 両辺 左から e_H^{-1} をかける

左 $= e_H^{-1} e_H e_H = e e_H = e_H$

右 $= e_H^{-1} e_H = e$ $\quad$ よって $e \in H$

$\forall a \in H$

$\exists b' \in H$

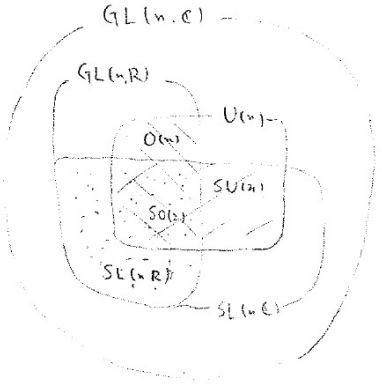
$bb' = b'b = e$

$b' = b'e = b'bb^{-1} = eb^{-1} = b^{-1}$

よって $b^{-1} \in H$

$\therefore ab^{-1} \in H$

1-4



$GL(n,\mathbb{C}) \supset GL(n,\mathbb{R}) \supset SL(n,\mathbb{R}) \supset O(n) \supset SO(n)$

$\det = 1$ となる a ; $\det \neq 0$ なもの

$GL(n,\mathbb{C}) \supset SL(n,\mathbb{C}) \supset SU(n)$

$\det = 1$

$SL(n,\mathbb{C}) \supset SL(n,\mathbb{R})$

$U(n) \supset O(n) \supset SO(n)$

Invariant subgroup $\left[\forall g \in G,\ gHg^{-1} \in H \ (= gH = Hg)\right]$

G : group $\supset H$: subgroup

$\forall h \in H \quad \forall g \in G \quad \underline{ghg^{-1}} \in H$

h の共役元

$g \in S_3 = \{e, (12), (13), (23), (123), (312)\}$

$H = \underset{\text{subgroup}}{S_2} = \{e, (12)\} \subset S_3$

$H = \{e\},\ H = G$

2つの trivial な

不変部分群」

$(123)e(123)^{-1} = e$

$(123)(12)(123)^{-1} = (23)$

$P = \begin{pmatrix} 1 & 2 & \cdots & n \\ P_1 & P_2 & \cdots & P_n \end{pmatrix}$

$P\,f(12)\,P^{-1}\,\psi(12\cdots n) = f(P_1 P_2)\,\psi(1,2,\cdots,n)$

semi-simple

simple

1-5

(i) $c \in C = \{c;\ gcg^{-1} = c,\ \forall g \in G\}$

$(gcg^{-1})^{-1} = c^{-1}$

(ii) $g = \begin{pmatrix} i & 0 \\ 0 & -i \end{pmatrix}$ $g^{-1} = \begin{pmatrix} -i & 0 \\ 0 & i \end{pmatrix}$ $g\begin{pmatrix} a & b \\ c & d \end{pmatrix}g^{-1} = \begin{pmatrix} a & -b \\ -c & d \end{pmatrix}$ $b=c=0$

$g = \begin{pmatrix} 0 & i \\ i & 0 \end{pmatrix}$ $g^{-1} = \begin{pmatrix} 0 & -i \\ -i & 0 \end{pmatrix}$ $g\begin{pmatrix} a & b \\ c & d \end{pmatrix}g^{-1} = \begin{pmatrix} d & 0 \\ 0 & a \end{pmatrix}$ $a=d$ $C^{\dagger}C = \begin{pmatrix} a^* & 0 \\ 0 & a^* \end{pmatrix}\begin{pmatrix} a & 0 \\ 0 & a \end{pmatrix} = \begin{pmatrix} |a|^2 & 0 \\ 0 & |a|^2 \end{pmatrix} = 1$

$C = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \begin{pmatrix} -1 & 0 \\ 0 & -1 \end{pmatrix}$

(iii) $U(2)$ a center $\begin{pmatrix} e^{i\theta} & 0 \\ 0 & e^{i\theta} \end{pmatrix}$ 群論 is Invariant group

$\neq$ semi group

1-6

(i) O(N)

$A \in O(N)$

$A = (u_1 u_2 \cdots u_N)$

${}^tAA = 1 \Leftrightarrow u_i u_j = \delta_{ij}$

よって $|u_i| = 1$ のN次元ベクトル

N次元球面



$N(N-1) - \binom{N}{2} = \frac{N(N-1)}{2}$

U(N)

$\vec{v}_i^{\,*} \vec{v}_j = \delta_{ij}$

$\vec{v}_i = e^{i\delta_i} \vec{v}_i$

Sがcompact ⇒ U(N) is compact

$U = (u_{ij})$

$U^+U = 1 \quad \sum_{j=1}^{n} u_{ij}^* u_{kj} = \delta_{ik}$

$u_{ij} = x_{ij} + i y_{ij} \quad \sum_{j=1}^{n} |u_{ij}|^2 = 1$

$\sum_{j=1}^{n} (x_{ij}^2 + y_{ij}^2) = 1$

(ii) $SL(2,C) \ni A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$

$ad - bc = 1$ より a,b,c,d は独立でない

$SL(2,\mathbb{C}) \ni S = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \quad a,b,c,d \in \mathbb{C}$

$\det S = ad - bc = 1$

$$S = a_0 + i \sum_{k=1}^{3} a_k \sigma_k = a_0 \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} + i a_1 \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} + i a_2 \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} + i a_3 \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

$$= \begin{pmatrix} a_0 + i a_3 & i(a_1 - i a_2) \\ i(a_1 + i a_2) & a_0 - i a_3 \end{pmatrix}$$

$$\det S = (a_0 + i a_3)(a_0 - i a_3) + (a_1 + i a_2)(a_1 - i a_2)$$

$$= a_0^2 + a_1^2 + a_2^2 + a_3^2 = \sum_{\mu=0}^{3} a_\mu^2 = 1$$

$a_\mu = x_\mu + i y_\mu$ と置き換え $(x_\mu, y_\mu \in \mathbb{R})$

$$\sum_{\mu=0}^{3} (x_\mu^2 - y_\mu^2) + 2i \sum_{\mu=0}^{3} x_\mu y_\mu = 1$$

$$\begin{cases} \sum_{\mu=0}^{3} (x_\mu^2 - y_\mu^2) = 1 \\ \sum_{\mu=0}^{3} x_\mu y_\mu = 0 \end{cases}$$

$$\begin{cases} x_\mu = \cosh \xi \hat{x}_\mu \\ y_\mu = \sinh \xi \hat{y}_\mu \end{cases} \quad \xi \in \mathbb{R}$$

$$\sum_{\mu=0}^{3} \hat{x}_\mu^2 = \sum_{\mu=0}^{3} \hat{y}_\mu^2 = 1$$

$$\sum_{\mu=0}^{3} \hat{x}_\mu \hat{y}_\mu = 0$$

$\hat{x}_\mu, \hat{y}_\mu$ 4×2=8

8−

1-(ii)

(iii) の後半

$\forall u \in SU(2) \quad \exists a_k \in \mathbb{R} \ (k=0\sim3) \quad \sum_{k=0}^{3} a_k^2 = 1$

$u = a_0 + i\sum_{k=1}^{3} a_k\sigma_k = \sum_{k=0}^{3} a_k\sigma_k \qquad \sigma_0 = 1$

$u = a_0\begin{pmatrix}1&0\\0&1\end{pmatrix} + ia_1\begin{pmatrix}0&1\\1&0\end{pmatrix} + ia_2\begin{pmatrix}0&-i\\i&0\end{pmatrix} + ia_3\begin{pmatrix}1&0\\0&-1\end{pmatrix}$

$= \begin{pmatrix} a_0+ia_3 & a_2+ia_1 \\ -a_2+ia_1 & a_0-ia_3 \end{pmatrix}$

$\det u = a_0^2 + a_3^2 - (ia_1)^2 + a_2^2 = \sum a_k^2 = 1$

${}^t u^* u = (a_0 - i\sum a_k\sigma_k^+)(a_0 + i\sum a_k\sigma_k)$

$= a_0^2 + \sum\sum a_k a_\ell \underbrace{\sigma_k\sigma_\ell}_{\delta_{k\ell} + i\epsilon_{k\ell j}\sigma_j}$

$= a_0^2 + \sum a_k^2 = 1 \quad \to u$ は unitary

(i) a を $a^{(\pm)}$ に分解 $\quad a = a_0 + ia_k I_k$ (real 4×4 matrix) $\qquad I_0 = 1$

${}^t a a = {}^t a^* a = 1 \quad \to a$ は直交行列 $(\det a^2 = 1)$

$\det a = -1$ は $O(4)$ のパラメータの連続性

${}^t(a^{(+)}a^{(-)})(a^{(+)}a^{(-)}) = {}^t a^{(-)}\underbrace{{}^t a^{(+)} a^{(+)}}_{1} a^{(-)} = 1$ より直交行列

問題は $SO(4)$ 全体をつくしているか？

パラメータの数 $2\times(4-1) = 6$

$\dim O(4) = \frac{4(4-1)}{2} = 6$ } O.K.

(ii) 直積の形

(iii) $T(a^{(+)}a^{(-)})T^{-1} = \underbrace{T a^{(+)} T^{-1}}\cdot\underbrace{T a^{(-)} T^{-1}}$

$a_k^{(+)}\otimes 1 + ia_k^{(+)}\sigma_k\otimes 1$

$= \left((a_0^{(+)} + ia_k^{(+)})\otimes 1\right)\left(1\otimes(a_0^{(-)} + ia_s^{(-)}\sigma_s)^*\right)$

$= \underbrace{(a_0^{(+)} + ia_k^{(+)})}_{SU(2)} \otimes \underbrace{(a_0^{(-)} + ia_s^{(-)}\sigma_s)^*}_{SU(2)}$

§2 有限次元

vector space　基底 $\{e_i\}\ i=1\sim n$

$X=\sum X_i e_i$

$Y=\sum Y_j e_j$

$[X,Y]=\sum\sum X_i Y_j [e_i,e_j] = \sum\left(\sum\sum X_i Y_j c_{ijk}\right)e_k$

$[e_i,e_j]=\sum c_{ijk}e_k$

T

構造定数

2-4

subalgebra　Lie環 ($\leftrightarrow$ Lie群 G)

$h \subset g$

$\forall X\in h\ \forall Y\in g \quad [X,Y]\in h$

Lie群の場合

$H\subset G$

$g^{-1}hg^{-1}\in H$

$gHg^{-1}\subset H \quad \text{for}\ \forall g\in G$

$h=e^{\varepsilon X}$

$g=e^{\varepsilon Y}$　ε: real　$|\varepsilon|\ll 1$

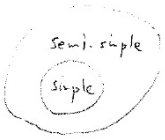
$$ghg^{-1}=e^{\varepsilon Y}e^{\varepsilon' X}e^{-\varepsilon Y}=1+\varepsilon' e^{\varepsilon Y}Xe^{-\varepsilon Y}=1+\varepsilon' X+\varepsilon\varepsilon'[Y,X]+\cdots$$

$$=e^{\varepsilon'[X,Y]\varepsilon}=e^{\varepsilon'[X,Y]\varepsilon}$$

2-5

Lie環 g が 0 と g 以外の ideal を持たない $\equiv$ g: simple Lie algebra

〃　単純な ideal 〃 $\equiv$ g: semi-simple



2-3

$SU(2)$

$\{e_x e_y e_z\}$

semi-simple であるとき 持つ

1次元 or 2次元 の 変部分 Lie環 が ideal

もし2次元なら $[e_x e_y]\to[e_x e_y]=e_z$ を $[e_x e_y]$

ダメ

1次元 $[e_x e_x]=0$ を $\{e_x\}$ がダメ

$=$ 任意

2-4

(i) $SU(2) \ni g \rightarrow \det g = 1 \ , \ gg^\dagger = 1$

① $g = e^{\epsilon X} \quad \det g = e^{\mathrm{Tr}\,\epsilon X} = 1 \rightarrow \mathrm{Tr}\,X = 0$

② $e^{\epsilon X^\dagger} e^{\epsilon X} = 1 + \epsilon(X + X^\dagger) + O(\epsilon^2) = 1 \quad X$: anti hermite

$$su(2) = \{X \mid X + X^\dagger = 0 \ , \ \mathrm{Tr}\,X = 0\}$$

(ii) $\{e_i\} = \left\{\frac{\sigma_i}{2i}\right\}$

$$X = \begin{pmatrix} ia_3 & -a_2 + ia_1 \\ a_2 + ia_1 & -ia_3 \end{pmatrix} = i\sum_k a_k \sigma_k$$

2-5 $\quad X = \begin{pmatrix} id_1 & ia & ib \\ i\bar{a} & id_2 & ic \\ i\bar{b} & i\bar{c} & id_3 \end{pmatrix}$ $\quad a,b,c \in \mathbb{C}$, $d_1 d_2 d_3 \in \mathbb{R}$ $\quad \mathrm{Tr}\,X = 0 \rightarrow d_1 + d_2 + d_3 = 0$

$3 \times 2 + 3 - 1 = 8$ (abc, d, Tr)

$$(e_i, e_j) = (-2)\,\mathrm{Tr}\,(e_i e_j) = (-2)\frac{1}{2i}\frac{1}{2i}\mathrm{Tr}\,(\sigma_i \sigma_j) = \delta_{ij}$$

$$0 = \mathrm{Tr}\left[\sigma_i \left\{\sum_j (X_j + X_j^*)\sigma_j\right\}\right]$$

$$= \sum_j (X_j + X_j^*)\,\mathrm{Tr}\,(\sigma_i \sigma_j) = 2(X_i + X_i^*)$$

$= \delta_{ij}$

2-6

$O(n), SO(n)$; $g = e^{\epsilon X} = 1 + \epsilon X + O(\epsilon^2)$ $\quad {}^t g g = 1 \rightarrow X + {}^t X = 0$

${}^t g = e^{\epsilon {}^t X} = 1 + \epsilon {}^t X + O(\epsilon^2)$

$\dim = n + (n-1) + \cdots + 1 = \frac{n(n-1)}{2}$

$U(n)$; $X^\dagger + X = 0$ $\quad \dim \ n + n(n-1) = n^2$

$SU(n)$; $X^\dagger + X = 0$, $\mathrm{Tr} X = 0$ $\quad \dim \ n^2 - 1$

$SL(n)$; $\mathrm{Tr} X = 0$ $\quad \dim \ 2n^2 - 2$

ϵ は real

$\mathrm{Tr} X$ の Re と Im

$SU(2)$ で $\frac{\sigma_k}{2i}$ を base にとる と 構造定数 は ϵ_{ijk}

$SO(3)$ $\quad X_1 = \begin{pmatrix} & & \\ & & -1 \\ & 1 & \end{pmatrix}$ $\quad X_2 = \begin{pmatrix} & & -1 \\ & & \\ 1 & & \end{pmatrix}$ $\quad X_3 = \begin{pmatrix} & -1 & \\ 1 & & \\ & & \end{pmatrix}$ $\quad$ これ を 構造定数 に すると $-\epsilon_{ijk}$

$$\begin{pmatrix} 0 & a & b \\ -a & 0 & c \\ -b & -c & 0 \end{pmatrix}$$

$(E_i)_{\alpha\beta} = \epsilon_{i\beta\alpha}$ $\quad E_1 = \begin{pmatrix} & & \\ & & -1 \\ & 1 & \end{pmatrix}$

$E_2 = \begin{pmatrix} & & 1 \\ & & \\ -1 & & \end{pmatrix}$

$E_3 = \begin{pmatrix} & -1 & \\ 1 & & \\ & & \end{pmatrix}$

$[E_i, E_j] = \epsilon_{ijk} E_k$ $\quad SO(3)$

$\updownarrow$ 同型 $\quad SI$

$[e_i e_j] = \epsilon_{ijk} e_k$ $\quad SU(2)$

局所同型

Lie 環 $\quad$ 群

$so(3) \longleftrightarrow SO(3)$

1:1 同型 $\quad$ 1:1 局所同型

$su(2) \longleftrightarrow SU(2)$

spin 1 の generator

2-7

(i) $su(3) \ni X$ は $X^\dagger + X = 0$ (anti hermite) $\mathrm{tr}\, X = 0$ (traceless)

3×3 行列の対角成分 pure imaginery かつ traceless → Λ_3 と Λ_8 を基底にとる

非対角成分は $X^\dagger + X = 0$ → $(\Lambda_1\ \Lambda_2)$, $(\Lambda_4\ \Lambda_5)$, $(\Lambda_6\ \Lambda_7)$ を基底

Trace 内積 $\mathrm{Tr}(\lambda_k^2) = 2$ $(\Lambda_k\ \Lambda_k) = 1$ (規格)

直交性 $\lambda_i \lambda_j$ が対角項をもつのは

$(i\ j) = (1\,2)(4\,5)(6\,7)(3\,8)$ の4個

(ii) (i,j) 両係数基底を $2i$ 倍,2 割る.

非対角成分については σ_1 と σ_2 の2つをそれぞれの成分の所に分解 ::: ::: く

対角成分は

$$\begin{pmatrix} 1 & & & \\ & -1 & & \\ & & 0 & \\ & & & \ddots \end{pmatrix} \quad \frac{1}{\sqrt{3}}\begin{pmatrix} 1 & & & \\ & 1 & & \\ & & -2 & \\ & & & \ddots \end{pmatrix} \quad \frac{1}{\sqrt{6}}\begin{pmatrix} 1 & & & \\ & 1 & & \\ & & -3 & \\ & & & \ddots \end{pmatrix} \quad \cdots \quad \frac{\sqrt{2}}{\sqrt{n(n-1)}}\begin{pmatrix} 1 & & & \\ & \ddots & & \\ & & 1-n & \end{pmatrix}$$

Gell-Mann matrix

$su(2) = \{X + {}^tX = 0,\ \mathrm{Tr}\, X = 0\}$

$X \in GL(3,\mathbb{C})$

$X \to \Lambda_i = \frac{\lambda_i}{2i}$

$\begin{cases} \lambda_i^\dagger = \lambda_i \\ \mathrm{Tr}\,\lambda_i = 0 \end{cases}$

$\mathrm{Tr}\,\lambda_i \lambda_j = 2\delta_{ij}$

$\mathrm{Tr}\, AB = A_{ij} B_{ji}$

$= A_{ij} B^*_{ij}$

$\lambda_1 = \begin{pmatrix} \sigma_1 & \\ & 0 \end{pmatrix} = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$ $\lambda_2 = \begin{pmatrix} \sigma_2 & \\ & \end{pmatrix}$ $\lambda_3 = \begin{pmatrix} \sigma_3 & \\ & \end{pmatrix}$

$\lambda_4 = \begin{pmatrix} 0 & 1 & \\ 1 & 0 & \\ & & \end{pmatrix}$ $\lambda_5 = \begin{pmatrix} & -i & \\ i & & \\ & & \end{pmatrix}$ $\widehat{\lambda_6} = \begin{pmatrix} 1 & & \\ & -1 & \\ & & \end{pmatrix}$

$\lambda_6 = \begin{pmatrix} & & \\ & 0 & 1 \\ & 1 & \end{pmatrix}$ $\lambda_7 = \begin{pmatrix} & & \\ & & -i \\ & i & \end{pmatrix}$ $\widehat{\lambda_8} = \begin{pmatrix} 1 & & \\ & -1 & \\ & & \end{pmatrix}$

$\lambda_3 + 2\widetilde{\lambda_8} = \begin{pmatrix} 1 & & \\ & 1 & \\ & & -2 \end{pmatrix}$

$\lambda_8 = \frac{1}{\sqrt{3}}(\lambda_3 + 2\widetilde{\lambda_8}) = \frac{1}{\sqrt{3}}\begin{pmatrix} 1 & & \\ & 1 & \\ & & -2 \end{pmatrix}$

$X_1^{\prime} = \begin{pmatrix} 1 & -1 & \\ & & \end{pmatrix}$ $X_2^{\prime} = \frac{1}{\sqrt{3}}\begin{pmatrix} 1 & & \\ & 1 & \\ & & -2 \end{pmatrix}$

$\lambda_1^R = \begin{pmatrix} 0 & 1 & \\ 1 & 0 & \\ & & \end{pmatrix}$ $\lambda_2^R = \begin{pmatrix} & 1 & \\ 1 & & \\ & & \end{pmatrix}$ $\lambda_3^R = \begin{pmatrix} 0 & 1 & \\ 1 & 0 & \\ & & \end{pmatrix}$

$\lambda_1^I = \begin{pmatrix} 0 & -i & \\ i & 0 & \\ & & \end{pmatrix}$ $\lambda_2^I = \begin{pmatrix} & -i & \\ i & & \\ & & \end{pmatrix}$ $\lambda_3^I = \begin{pmatrix} 0 & -i & \\ i & 0 & \\ & & \end{pmatrix}$

2-17 つづき

SU(n) の generator　　Cartan subalgebra

$$\lambda^d_k = \sqrt{\frac{2}{k(k-1)}} \begin{pmatrix} 1 & & & & \\ & \ddots & & & \\ & & 1 & & \\ & & & -k & \\ & & & & 0 \\ & & & & & \ddots \\ & & & & & & 0 \end{pmatrix} \quad n-1$$

$k = 1 \sim (n-1)$

$$(i<j) \quad \lambda^R_{ij} = \begin{array}{c} \\ i \\ \\ j \\ \\ \end{array} \begin{array}{c} \begin{matrix} & & i & & j & \\ \end{matrix} \\ \begin{pmatrix} \cdots & 0 & 1 & \cdots \\ \vdots & \vdots & \vdots & \vdots \\ \cdots & 1 & 0 & \cdots \\ & & & \ddots \\ & & & & 0 \end{pmatrix} \end{array} \quad \frac{n(n-1)}{2}$$

$$\lambda^2_{ij} = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \quad \frac{n(n-1)}{2}$$

n^2-1

2/18

7-(9) $\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$ · symplectic matrix　　$\overset{p}{L_Q}$　　$\begin{pmatrix} \dot{Q} \\ \dot{p} \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}\begin{pmatrix} Q \\ p \end{pmatrix}$

$\dot{p} = -Q = -\frac{\partial V(Q)}{\partial Q}$

$V(Q) = \frac{1}{2}Q^2$

1-7

(I) $(\alpha, \beta) = {}^t\alpha g \beta$　　a, b は (I) の内積不変に対する一次変換

$(\alpha', \beta') = (a\alpha, a\beta) = {}^t(a\alpha) g (a\beta) = {}^t\alpha\, {}^t a g a \beta$　　$g = {}^t a g a$

$g = {}^t b g b$

${}^t(ab) g (ab) = {}^t b ({}^t a g a) b = g$ より不変

$(b a \in G)$

$G = \{a \mid {}^t a g a = g,\ a \in GL(n, \mathbb{R})\}$

単位元　単位行列

逆元　逆行列

(II) t と t' について

1-8　(I) ${}^t O O = 1$　実直交行列

(II) $U^\dagger U = 1$　ユニタリー行列

1-9　$g = \begin{pmatrix} 0 & & \\ & \ddots & \\ & & 0 \end{pmatrix}$　$1+2+\cdots+(2n-1) = n(2n-1)$ 次　　Q は $2n$ 次

$({}^t g = -g)$